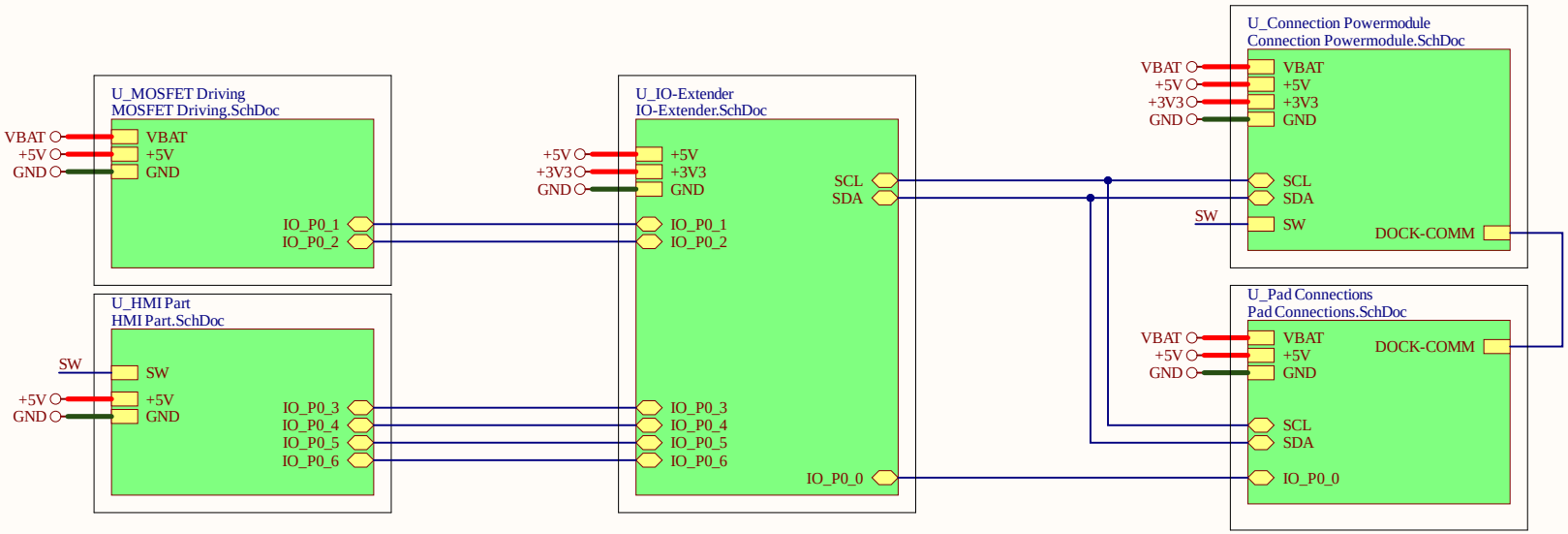
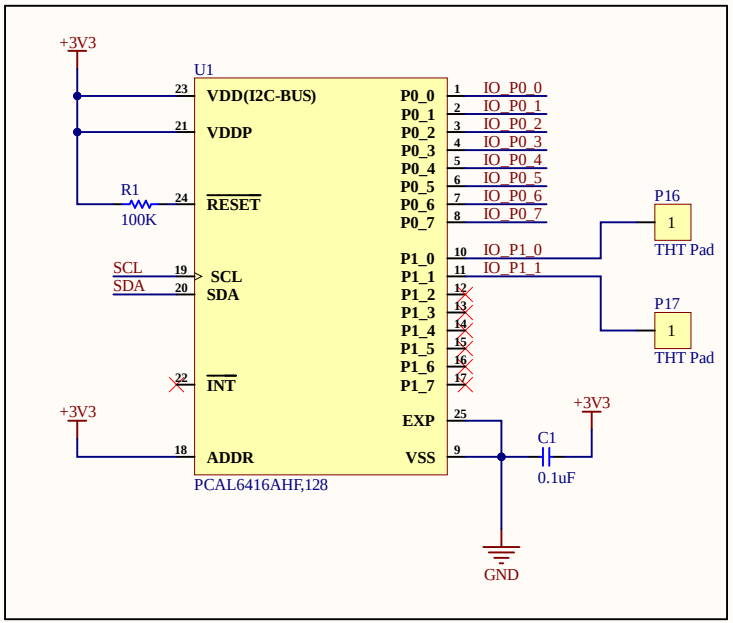


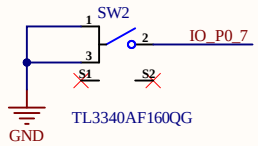
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D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

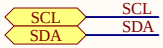
Title: Top-Sheet.SchDoc	
File: Smart Power Supply Unit-v4.0.0.PrjPcb	
Date: 2/22/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 6

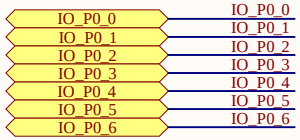
IO_EXPANDER

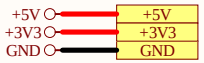


Internal pullup/pulldown resistors within IO Expander



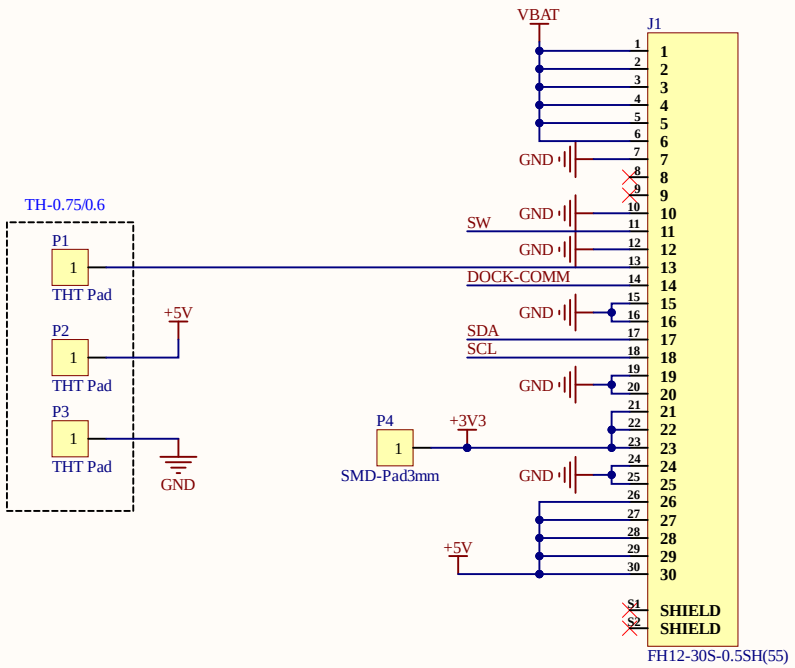




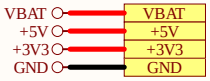


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Title: IO-Extender.SchDoc	
File: Smart Power Supply Unit-v4.0.0.PrjPcb	
Date: 2/22/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 6



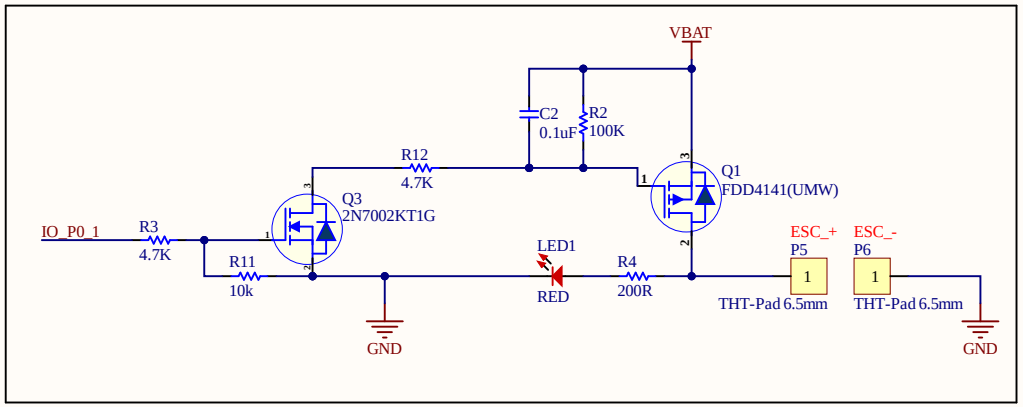




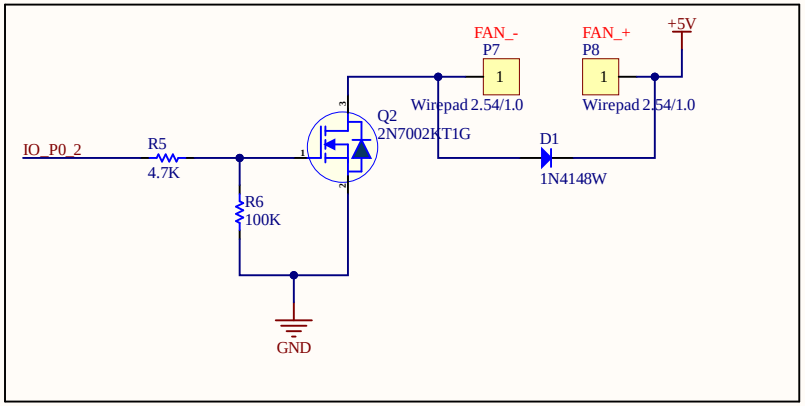
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Title: Connection Powermodule.SchDoc	
File: Smart Power Supply Unit-v4.0.0.PrjPcb	
Date: 2/22/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 6

MOSFET_ESC



MOSFET_FAN



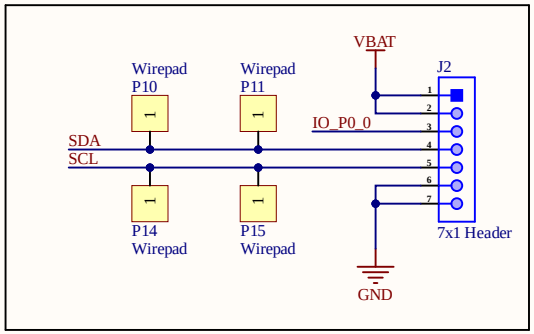
IO_P0_1
IO_P0_2
IO_P0_1
IO_P0_2

VBAT
+5V
GND
VBAT
+5V
GND

Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

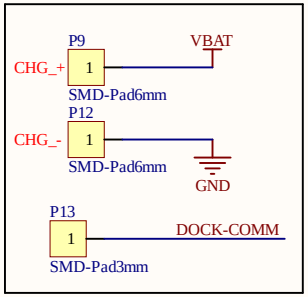
Title: MOSFET Driving.SchDoc	
File: Smart Power Supply Unit-v4.0.0.PrjPcb	
Date: 2/22/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 6

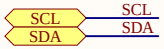
BATTERY_CONNECTOR



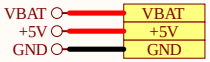
2S LiPo 8.4V (6V discharge voltage)

CHARGING_CONN



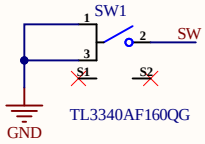


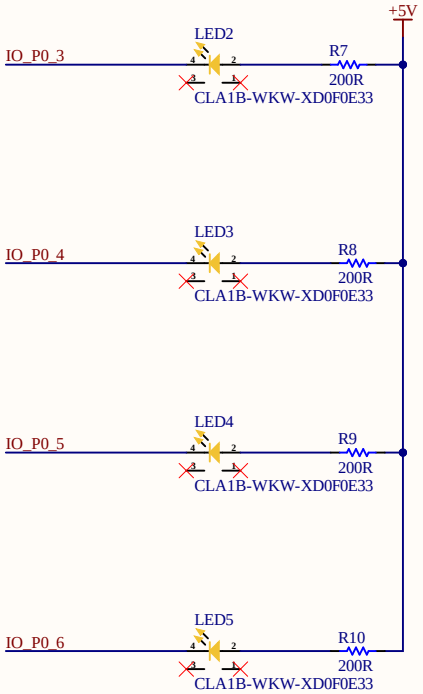


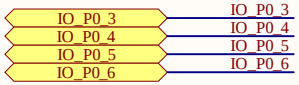


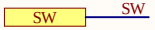
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Title: Pad Connections.SchDoc	
File: Smart Power Supply Unit-v4.0.0.PrjPcb	
Date: 2/22/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 6





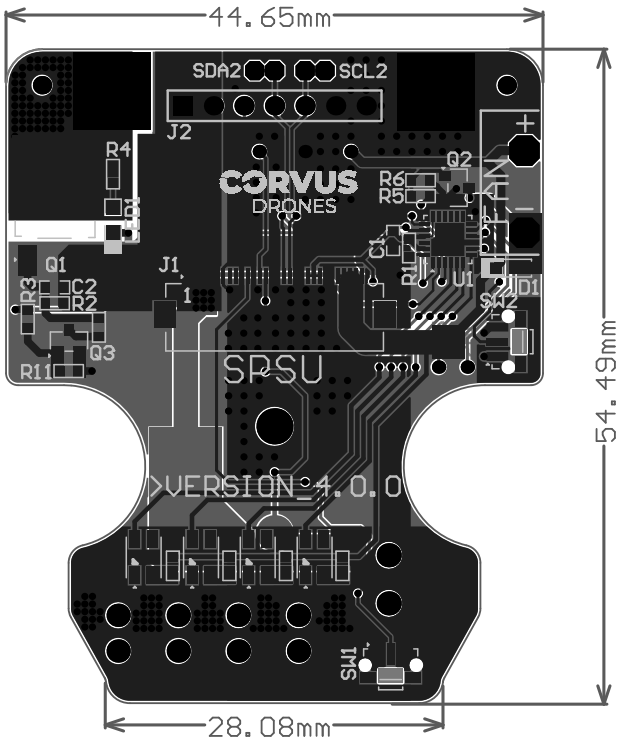






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Title: HMI Part.SchDoc	
File: Smart Power Supply Unit-v4.0.0.PrjPcb	
Date: 2/22/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 6



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.39mil	3.5	
1	Top Layer		1.38mil		
	Dielectric 1	FR-4	59.06mil	4.8	
2	Bottom Layer		1.38mil		
	Bottom Solder	Solder Resist	0.39mil	3.5	
	Bottom Overlay				
Total board thickness:			62.60mil		